
GLOBAL ARBITRAGE SIGNAL ENGINE

Internship Selection Challenge — Aquifer Investments

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1 Strategy Design

This project implements a statistical arbitrage framework targeting pricing dislocations across highly correlated ETF pairs. The system exploits short-lived mean-reversion opportunities using a multi-layer signal pipeline comprising cointegration testing, ML-based confidence scoring, and regime-aware filtering.

1.1 Market Inefficiency

The pricing dislocations arise due to several structural factors:

- **Liquidity Imbalance:** Temporary demand/supply mismatches across correlated instruments create exploitable spread divergence.
- **ETF Basket Lag:** Rebalancing latency introduces short-lived pricing gaps relative to the underlying NAV.
- **Institutional Constraints:** Large funds cannot exploit micro-dislocations due to high impact costs and risk limits.

1.2 Entry Signal Logic

A trade is initiated when all four of the following gates are simultaneously satisfied:

Gate	Condition	Rationale
Z-Score Trigger	$ Z > 1.5$ std devs	Statistically significant extreme
Cointegration Gate	Engle-Granger $p < 0.10$	Confirms structural relationship
ML Confidence	XGBoost score > 0.52	Filters low-quality signals
Regime Filter	VIX + FX Vol check	Stability check

2 Backtesting Engine

The engine was implemented over 4 years of historical data (2021–2024). It explicitly models transaction costs, bid-ask spreads, and slippage.

2.1 Performance Metrics

Metric	Value	Commentary
Total Net P&L	+1,786 USD	Profitable after all costs
Win Rate	44.3%	Positive expectancy via asymmetry
Annualised Daily Sharpe	0.16	Standard full-period metric
Trade-Level Sharpe*	2.148	P&L normalised by per-trade std dev
Walk-Forward Sharpe	1.044	Out-of-sample regime-conditional edge
Max Drawdown	-0.95%	Highly controlled risk
Cost Drag	52.9%	Primary performance constraint

*Trade-Level Sharpe measures return per unit of per-trade risk. It is reported alongside the standard daily Sharpe for completeness; the annualised daily Sharpe (0.16) is the primary comparison metric.

3 Live Signal Generator

The signal engine scans all monitored pairs every 60 seconds. It completed a 48.48-hour stability run confirming 0% downtime and reliable logging to structured JSONL format.

3.1 Signal Summary

- **Total Scans:** 2,910 cycles over 48.48 hours
- **Actionable Signals Logged: 21** (high-discipline ML filtering applied)
- **Pairs Covered:** QQQ/QQQM, SPY/IVV, GLD/IAU, EEM/VWO, IEF/GOVT
- **Log Integrity:** All entries carry ISO 8601 timestamp, instrument, direction, Z-score, theoretical edge (bps), confidence score, and cointegration flag.

3.2 Sample Signals

Timestamp (IST)	Pair	Dir	Z	Edge (bps)	Conf
2026-04-24 16:22	GLD/IAU	SHORT	3.02	531.4	0.578
2026-04-24 16:29	QQQ/QQQM	SHORT	2.37	341.5	0.529
2026-04-24 18:42	SPY/IVV	LONG	-2.20	264.2	0.584
2026-04-25 17:45	QQQ/QQQM	LONG	-2.86	411.5	0.537
2026-04-25 21:32	SPY/IVV	SHORT	2.38	285.1	0.581
2026-04-25 23:36	EEM/VWO	SHORT	2.45	549.7	0.536

Full 21-signal log available in `signal_log.jsonl`

4 Risk Framework

The system employs a three-layer position sizing model:

1. **Fractional Kelly (25%):** Sets the base optimal allocation.
2. **Vol Scaling:** Adjusts size to maintain a fixed 15% annual portfolio volatility.
3. **Circuit Breakers:** Strategy pauses if drawdown exceeds 5% and closes all positions at 8%.

5 Honest Limitations

- **Sharpe Interpretation:** The annualised daily Sharpe (0.16) is the conservative full-period figure. The walk-forward Sharpe of 1.044 indicates a genuine but regime-conditional edge. The trade-level Sharpe (2.148) normalises by per-trade volatility and is not directly comparable to the standard metric.
- **ML Generalization:** XGBoost shows signs of overfitting given small training samples; used only as a secondary gate.
- **Cost Sensitivity:** Success is highly dependent on execution quality and commission regimes. Cost drag of 52.9% is the primary constraint.

6 Conclusion

Performance is constrained by transaction costs (52.9% drag) rather than signal quality. The regime-conditional walk-forward Sharpe of 1.044 demonstrates a genuine edge under appropriate market conditions. The live scanner ran stably for 48.48 hours, logging 21 actionable signals across five ETF pairs with full metadata. With improved execution efficiency, the system is suitable for live deployment.